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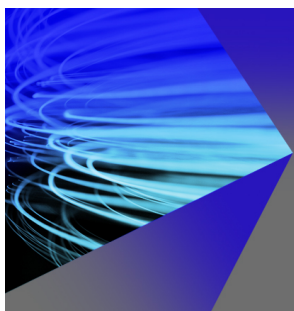
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Stability and Structure of Burner Flames^{1, *}

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It has been shown that the condition of equality of gas and burning velocities, required for stabilization of a flame above a burner, is established near the rim of the orifice or an obstruction within the stream by the effects of friction and inhibition of the explosive reaction. This condition is maintained between two critical gradients of the gas velocity at the solid surface, the lower gradient bordering on the flash-back and the upper gradient on the blow-off range. Values of the gradients in the range of laminar flow were determined by hydrodynamic equations from gas flow and tube dimensions; and their independence of tube diameter, except for extreme sizes, has been demonstrated both for upright and inverted flames. In the latter the critical velocity gradient for blow-off was also found to be independent of the diameter of the centrally mounted wire within a considerable range. The effect of the surrounding atmosphere on the critical blow-off gradient has been shown. The gas-flow pattern was studied experimentally by photographing stroboscopically illuminated magnesium oxide dust particles. No appreciable redistribution of velocities over the cross section of the stream was observed below the combustion zone. The burning velocity was found constant over the surface of the inner cone,

except at the tip, where it increases to the axial gas velocity, and at the base, where it decreases to zero. The experimental flame cone outline and flow pattern agree with the theoretical within the limitations imposed by simplifying assumptions. The temperature distribution in the flame was determined by the sodium line-reversal method, only the center of the flame being colored, and observations of the width and emission spectrum of the luminous combustion zone were made. In natural gas-air flames C-C and C-H bands are observed in the zone, and the temperature rises gradually behind the zone to a maximum that corresponds to the theoretical flame temperature. In natural gas-oxygen flames no C-C or C-H bands are observed; and the temperature attains a maximum, exceeding the theoretical, immediately behind the combustion zone. Temperature distribution and flow pattern have been correlated. The flow pattern of inverted flames was also studied with particular attention to the region of flame attachment just above the central wire where the formation of an annular vortex is demonstrated. An interpretation of the formation of polyhedral flame cones, described by Smith and Pickering, has been given.

WHEN an explosive gas mixture is ignited, combustion takes place in a narrow zone, separating burned and unburned gas, which propagates as a Huygens wave in a direction normal to itself into the unburned gas at the rate at which combustion reactions are induced in successive gas layers by the processes of heat transfer (essentially conduction) and diffusion of active species. The rate of progress of the combustion zone is called the burning velocity. A burner flame remains stationary above the orifice because there are regions, either near the rim or near an obstruction in the stream like the grid of a Méker burner, where the gas velocity equals the burning velocity. These regions serve

as continuous sources of ignition for the neighboring gas elements whose velocity exceeds the burning velocity. As the combustion zone propagates from these regions of equality it assumes an angle to the direction of gas flow so that everywhere the normal component of the gas velocity equals the burning velocity. For example, in the ordinary Bunsen burner flame the region of equality is established in a concentric circle above the orifice, and the combustion zone must take the form of an upright, circular cone.

The theory of burner flames has been comprehensively discussed by Mache.⁴ His conclusions, however, do not explain satisfactorily how the regions of equality of gas velocity and burning velocity are established⁵ and consequently do not yield the exact conditions for flash-back and blow-off. This omission is reme-

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^{*} Presented at the Buffalo meeting of the American Chemical Society.

⁴ H. Mache, *Die Physik der Verbrennungerscheinungen* (Leipzig, 1918).

⁵ Cf. W. Jost, *Explosions- und Verbrennungsvorgänge in Gasen* (Berlin, 1939), pp. 72 et. seq.



FIG. 1. Tracks of stroboscopically illuminated magnesium oxide particles in a flame burning on a rectangular tube of 0.755 cm $\times$ 2.19 cm cross section. The short side of the burner tube faces the camera. The light beam, about 1 mm wide and several cm high, bisects the long side. Time between interruptions, 1.44×10^{-3} sec. Mixture composition 7.50 percent natural gas in air; gas flow, 204 cc/sec.

died in the present paper. Furthermore, the fundamental concepts of the theory of flame structure have been verified and extended in detail by experimental studies of the direction and velocity of the gas flow (flow pattern) throughout the flame, of the burning velocity in various parts of the combustion zone, and of the distribution of temperature, so that a comprehensive description of burner flames can now be given that should be applicable to various problems in gas technology.

1. APPARATUS AND PROCEDURE

Mixtures of natural gas and air were used in most of the experiments. Air was taken from a constant-pressure reservoir, and all other gases were drawn from tanks, using two-stage regulators to insure constant flow. The Orsat analysis of the natural gas corresponded to 81.8 percent methane, 17.7 percent ethane, and 0.5 percent nitrogen, and hence the stoichiometric percentage

in air was 8.49. The actual composition was somewhat different because of the presence of traces of higher hydrocarbons. The gas and air were passed separately through calibrated flowmeters to a mixing chamber. The mixture composition as determined from the flowmeter readings was confirmed by analysis. In determinations of blow-off and flash-back the mixture passed directly to Pyrex burner tubes of about 1-meter length to insure laminar flow at the orifice, of inside diameters from 0.315 to 1.550 cm and a wall thickness of about 1 mm. For blow-off experiments in atmospheres of gases other than air, the burner tube was surrounded by a wide glass tube through which a stream of the gas was passed upward around the flame. For the study of blow-off of inverted flames, a straight wire of steel or brass with a flat end was mounted in the axis of the tube and the end of the wire extended about 1 cm above the rim. The other end of the wire was looped into a spiral that fitted the tube snugly and served as a support at a distance of over 40 times the tube diameter below the rim. Gas mixtures were ignited by applying a small flame to the region of low gas velocity near the rim of the tube or, in experiments with inverted flames, just above the central wire. Between trials the burner was cooled to room temperature.

The gas-flow pattern was determined by photographing stroboscopically illuminated dust

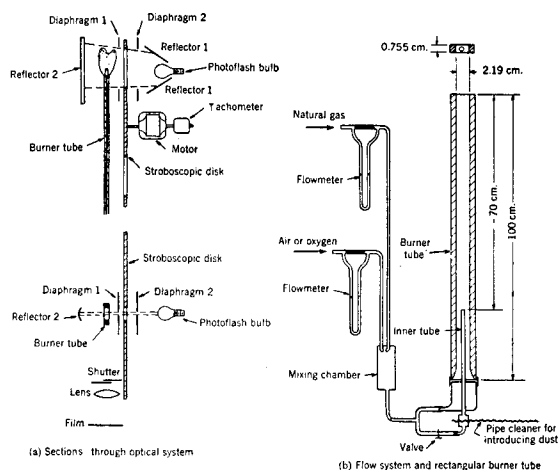


FIG. 2. Apparatus for determining the flow pattern of ignited and unignited gas streams.

particles carried by the stream.⁶ The photographs show the flame outline and interrupted particle tracks (Fig. 1), the velocity of the particles being determined from the distance between two interruptions and the speed of the stroboscopic disk. The experimental arrangement is shown in Fig. 2. Diaphragms 1 and 2 define a 1 to 2-mm wide sheet-like beam of light from the photoflash bulb and reflectors 1, which is interrupted by the rotating disk. The beam passes across the center of the orifice of the burner and is reflected in itself by cylindrical reflector 2; thus the dust particles are illuminated from two sides. The camera is arranged at right angles to the direction of illumination. The stroboscopic disk was enclosed in a housing to protect the flame from drafts and was divided into 24 sectors open over four-fifths of their width. Except as otherwise indicated experiments were made with the disk rotating at 1740 r.p.m., as measured by a tachometer. The photographic shutter consisted of four blades which opened and closed completely in 0.05 second with maximum opening about 40 percent of the time, and carried electric contacts to synchronize photoflash and photographic exposure. The photoflash bulbs were Mazda No. 21 with a peak of $4\frac{1}{2}$ million lumens and more than $2\frac{1}{2}$ million lumens for 0.01 second. A $3\frac{3}{4}$ -inch diameter, $f:2.5$, Taylor-Hobson lens and triple-S pan film were used. For experiments with upright flames, a brass burner tube 1 meter long and of rectangular cross section ($0.755\text{ cm} \times 5\text{ cm}$) was used, the longer dimension being reduced to 2.19 cm by inserting two metal strips, as shown in Fig. 2.

MgO dust (metallographic polish) was used. The particles produce sharp interrupted tracks, as their incandescence in the burned gas is insufficient to register on the film. Talcum powder is also suitable but not aluminum dust, which burns brightly. The dust was added to the stream at the bottom of the tube by inserting a pipe cleaner, covered with the dust, through opposite holes in the tube and moving it back and forth. No appreciable leakage of gas occurred through the holes past the pipe cleaner. The

particles were distributed over the entire cross section, but only those in the path of the light beam registered on the film. It was possible to concentrate the particles by introducing them into a smaller tube mounted in the center of the burner tube, as shown in Fig. 2. The gas stream was divided before entering the tubes and the velocity regulated so that the stream was laminar in the inner and outer tubes. The particles remained centrally located during their travel up the burner tube (about 70 cm). This distance

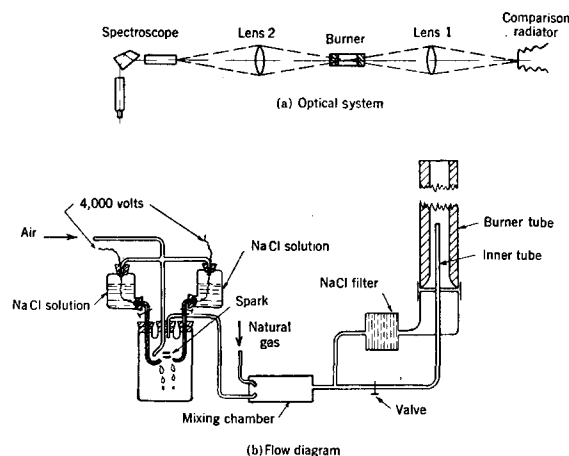


FIG. 3. Apparatus for determining temperatures in center of burner flame by the sodium line-reversal method.

sufficed to restore the normal velocity distribution over the cross section of the burner tube before reaching the orifice. The rectangular burner is advantageous compared to a cylindrical burner because in the flame no flow lines in the zone of illumination are refracted away from or toward the camera. The films were measured by mounting them on an accurately ruled grid and reading between the grid lines by means of a fine scale on a transparent film, specially drawn and photographed, mounted under a lens. The readings were transferred to coordinate paper, of which Fig. 24 is an example.

Some photographs were made with smoke substituted for particles; such smoke was produced by passing moist air over stannic chloride impregnated pumice, forming solid hydrates, probably mostly the trihydrate.

Flame temperatures were measured by the sodium line reversal method, which has been

⁶ F. A. Smith, Chem. Rev. 21, 389 (1937) used the method to determine velocity distribution in an unignited stream.

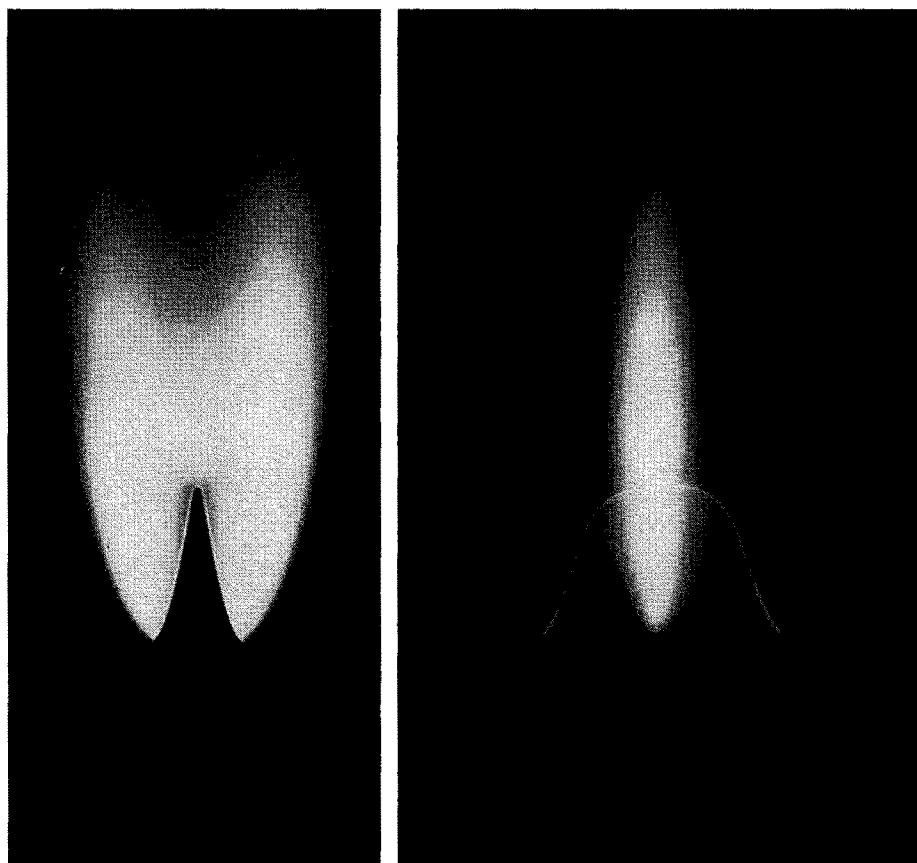


FIG. 4. Front and side views of flame on rectangular burner, colored centrally with sodium chloride.

described elsewhere.⁷ The apparatus was mounted on an optical bench and is shown diagrammatically in Fig. 3a. The image of the tungsten band comparison radiator is focused onto the center of the flame and both the former and the image of the flame are in turn focused on the slit of the spectroscope. To obtain correct temperature readings the light-gathering power of lens 1 must not be smaller than that of lens 2. The tungsten band lamp was calibrated at the Nela Park Laboratory of the General Electric Company for brightness temperature in the red as a function of current. Corrections were made for reflection from lens 1 and for yellow brightness as described elsewhere.⁸ Temperature readings could be reproduced within 4°C. The images of flame and

band on the slit were approximately of natural size; therefore, the temperature reading corresponded to the colored portion of the flame within a beam of the cross section of the slit, which was 0.02 mm wide and 0.1 mm long. To insure that measured temperatures corresponded to the center of the flame where the flow pattern was determined, only the central portion of the flame was colored by passing sodium chloride mist through the inner tube. The mist was formed in the air stream before entering the mixing chamber by continuously sparking between the open ends of two small glass tubes from which NaCl solution dripped (Fig. 3b). Continuous flow of fresh solution prevented sodium chloride from caking on the electrodes. Metal wires were introduced into the tubes to prevent the formation of bubbles by electrostriction. The bulk of the mixture was passed

⁷ B. Lewis and G. von Elbe, *Combustion, Flames, and Explosions of Gases* (Cambridge Press, 1938), pp. 333–337.

⁸ Jones, Lewis, Friauf, and Perrott, *J. Am. Chem. Soc.* **53**, 869 (1931).

through a cotton filter to remove the mist before it entered the main burner tube, while a small part of the mist-laden mixture passed directly into the inner tube. Figure 4 shows front and side views of the centrally colored flame. Some spreading of the sodium vapor takes place by diffusion. Although the flame was brightly colored, the concentration of NaCl was insufficient to lower the temperature appreciably. The entire flame could be explored by moving the burner tube vertically and horizontally while the rest of the system remained fixed; the point of measurement relative to the axis of the burner and the orifice was read on scales with Vernier attachments to 0.1 mm. 250 to 300 readings were made for the temperature contours in each of Figs. 31 to 34.

2. CONDITIONS FOR STABILITY OF FLAME ABOVE THE ORIFICE

(a) Upright Flames

Figure 5 is a photograph of a dust-laden laminar air stream (Reynolds number=1500) from a cylindrical tube 1.298 cm in diameter. The measured gas velocities near the orifice at various distances from the axis, corrected for free fall of

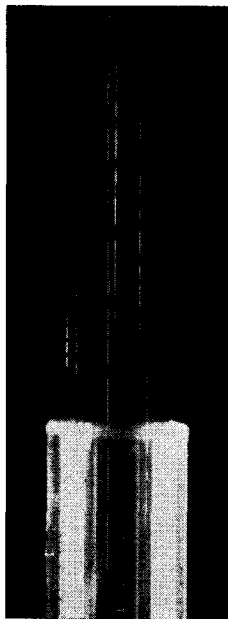


FIG. 5. Particle tracks in an air stream from a cylindrical tube of 1.298-cm diameter; gas flow, 241 cc/sec.; time between interruptions, 2.08×10^{-3} sec.

the particles (average 5 cm per second) are plotted in Fig. 6. They follow closely the Poiseuille parabola which is shown as a full line in the figure and is calculated from the equation:

$$U = n(R^2 - r^2). \quad (1)$$

U is the velocity at distance r from the axis, R is the radius of the tube, and $n = -a/4\eta$, where η is the viscosity and a the hydrodynamic pressure gradient along the length of the tube. The value of n is determined from flow V (volume per unit time) and radius R by the equation:

$$V = \int_0^R 2\pi U r dr = \frac{\pi}{2} n R^4. \quad (2)$$

The velocity distribution gradually changes downstream from the orifice owing to transfer of momentum from the inner to the outer flow lines and the surrounding atmosphere; it is noteworthy that, except for very small flow, the velocity in any flow line remains virtually con-

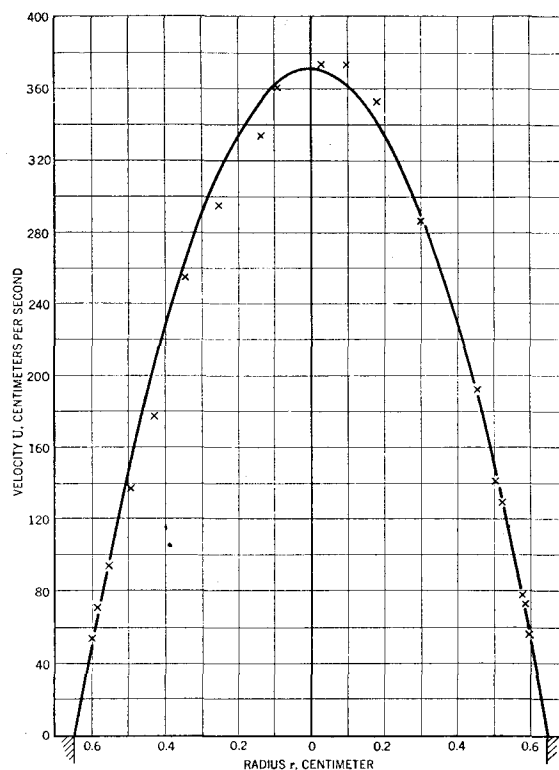


FIG. 6. Velocity distribution in a laminar stream of air (241 cc per second) at the orifice of a cylindrical tube of 1.298-cm diameter; temperature, 25°C; pressure, 740 mm Hg. x, experimental velocity. —, calculated velocity.

stant over a length of the order of the distance between orifice and combustion zone. That there is no noticeable slipping at the wall of the tube, as postulated by Poiseuille's equation, is confirmed by the experimental points near the boundary, since the line through these points would extrapolate to zero at the boundary. This result was confirmed in other determinations in different size tubes and for different flows.

The wall not only reduces the gas velocity by friction but also the burning velocity by its quenching effect on the explosive reaction. This effect is observed even in the usual kinetic experiments where the gas and wall are at the same temperature and only the destruction of chain carriers is involved. With a cold burner tube there is the additional effect of cooling, which is in itself very powerful because chain-carrier reactions generally require substantial activation energies. Hence, when, at flows in the flash-back range, a flame travels down the tube, it is observed that the luminous combustion zone does not make contact with the wall and that its fringe trails behind the more central portions; that is, the burning velocity is zero over a short distance from the wall and, farther in, gradually increases to its normal value for undisturbed explosive reaction as shown in Fig. 7a. Flash-back occurs if the burning velocity exceeds the gas velocity somewhere in the stream, as illustrated by curve 1 of Fig. 7a. As the flow is increased, a critical gradient of the gas velocity at the boundary is reached (curve 2) where flash-back is just possible. On further increase of the flow the gas velocity exceeds the burning velocity everywhere in the stream (curve 3), and the flame is swept out of the tube. The critical

velocity gradient for flash-back is obviously determined solely by factors that affect the burning velocity near the wall, namely, the condition of the wall and the mixture composition. Outside of the tube the gas near the boundary regains its explosivity; the combustion zone establishes itself as close to the rim of the orifice as the quenching effect on the explosive reaction permits. At some point gas and burning velocities are equal, as illustrated by curve 1 of Fig. 7b, and the flame is thus attached to the orifice. When the flow is increased, the combustion zone moves away from the rim and thus gains in burning velocity until a new point of equality is established farther out and farther above the rim. This cannot be continued indefinitely because, independent of the quenching action of the rim, the burning velocity vanishes toward the boundary of the stream; eventually (as illustrated by curves 2 and 3) the gas velocity exceeds the burning velocity everywhere in the stream, and the flame blows off the tube. Thus, there is a critical velocity gradient at the boundary for blow-off. The reason why the burning velocity vanishes toward the boundary needs no explanation if the surrounding atmosphere is a diluent gas that reduces the explosivity by interdiffusion. However, if the surrounding atmosphere is a gas that enriches the mixture, the following reasons must be considered: (1) The mixture is over-enriched in the outermost flow lines; (2) the gas movement extends beyond the confines of the explosive mixture by transfer of momentum. It is clear that, with diluent surrounding atmospheres, the critical gradient is much smaller than with enriching surrounding atmospheres.

The relation between the velocity gradient at the boundary,

$$\lim_{r \rightarrow R} \left(-\frac{dU}{dr} \right),$$

and experimental quantities V and R , is found by differentiating Eq. (1).

$$\lim_{r \rightarrow R} \left(-\frac{dU}{dr} \right) = 2nR. \quad (3)$$

Combining with Eq. (2) this becomes

$$\lim_{r \rightarrow R} \left(-\frac{dU}{dr} \right) = \frac{4}{\pi} \frac{V}{R^3}. \quad (4)$$

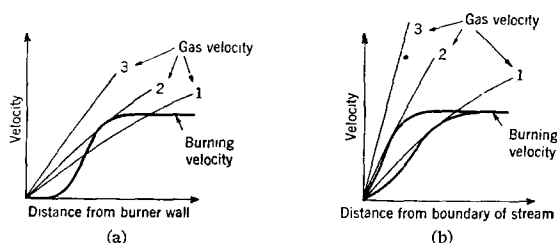


FIG. 7. Illustration of flash-back and blow-off conditions. (a) Velocity gradients at wall of burner tube. Curve 1, flash-back; curve 2, flash-back limit; curve 3, no flash-back. (b) Velocity gradients at boundary of stream above orifice. Curve 1, stable flame; curve 2, blow-off limit; curve 3, blow-off.

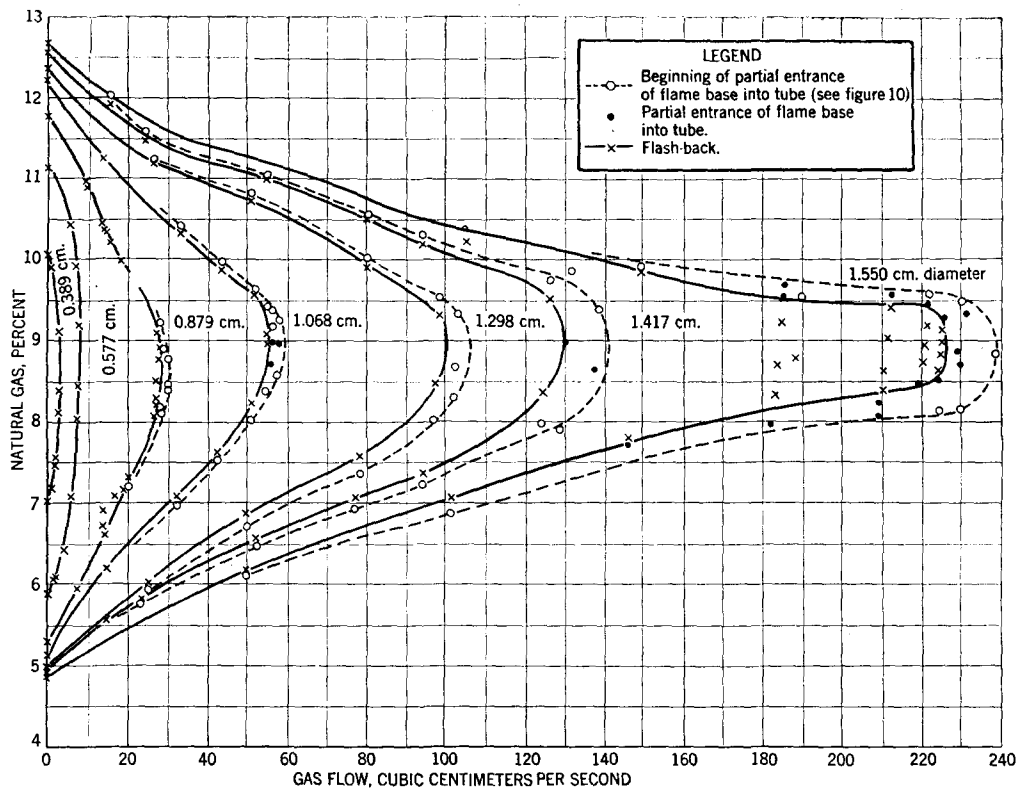


FIG. 8a. Critical flows for flash-back of natural gas-air flames in cylindrical tubes of different diameters at room temperature.

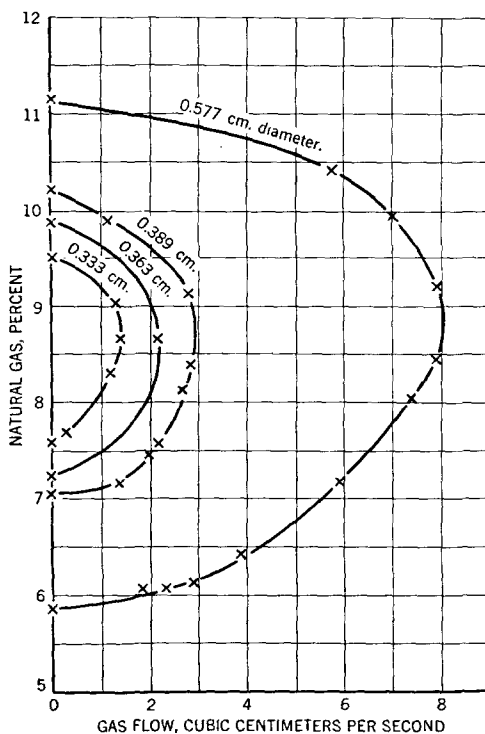


FIG. 8b. Critical flows for flash-back of natural gas-air flames in cylindrical tubes of different diameters at room temperature.

Figures 8a and 8b show critical flows, that is, maximum flows at which flash-back occurs, as a function of mixture composition in cylindrical glass tubes of different diameters at room temperature. The results are little affected by increase of tube temperature up to about 100°C . The points at zero flow, where the wall exerts its quenching effect on the explosive reaction over the whole cross section, are usually referred to as the limits of inflammability for downward propagation. These limits are plotted separately in Fig. 9. At flows just above the flash-back curves (solid curves, Fig. 8), flames of the type shown in Fig. 10, in which the base of the flame cone is partly drawn into the tube, are formed spontaneously. The formation of such flames may be explained as follows: If some point on the rim of the orifice becomes overheated, which may occur fortuitously by some small irregularity, the burning velocity at this point exceeds the gas velocity in the boundary layer and the flame penetrates into the tube at this point, the penetration gradually increasing unless arrested by cooling the tube. Because of confinement of the

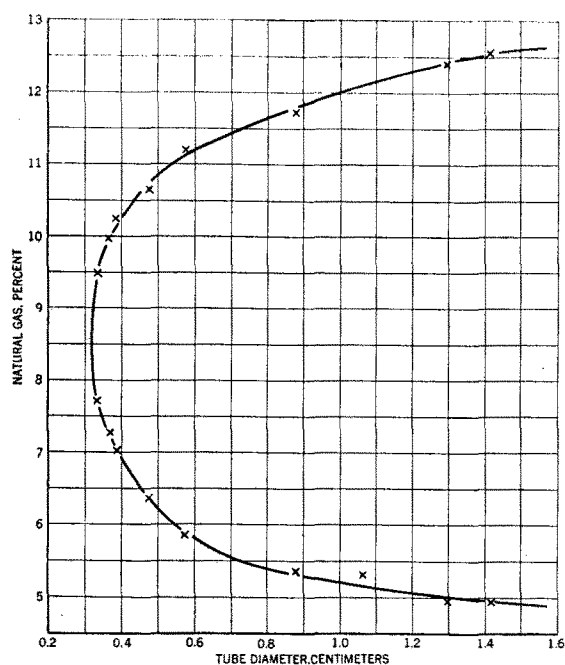


FIG. 9. Limits of inflammability in natural gas-air mixtures for downward propagation.

gas in the tube the expansion of the burned gas crowds the unburned gas to the side, increasing its velocity and keeping the flame there well above the orifice. The flow range for occurrence of flames of this type is confined to the region between dotted and solid curves in Fig. 8a. Above the flow indicated by the dotted curves normal flame cones are stable above the orifice, although over some additional range flames as in Fig. 10 may be forced by heating the tube locally.

Figure 11 is a plot of

$$\lim_{r \rightarrow R} \left(-\frac{dU}{dr} \right),$$

calculated from Eq. (4) for the flash-back curves in Figs. 8a and 8b, against mixture composition. Coincidence of the curves confirms that flash-back is essentially controlled by the velocity gradient at the wall. Exceptional deviations occur only in tubes close to the limiting diameter of 0.315 cm in which flame propagation in the most explosive mixture is arrested (see Fig. 9), although even with a tube diameter of 0.333 cm, close coincidence is found near the stoichiometric. The effect of tube diameter is further illustrated in Fig. 12. As the tube diameter is

decreased the critical boundary velocity gradient first increases and then decreases sharply to zero. The increase is caused by the fact that the branch of the parabola lying between the wall and the point of equality of burning velocity and gas velocity (Fig. 7a, curve 2) extends over a larger fraction of the radius and is, therefore, more strongly curved. When the tube diameter is decreased to the range where the burning velocity in the center is depressed below its normal value, the critical velocity parabolas again become flatter; this diameter range, in the case illustrated by Fig. 12, lies between 0.39 cm and 0.35 cm, so that the critical velocity gradient rapidly drops to zero. The deviation around the stoichiometric mixture for the largest tube (1.550 cm) may be linked to the pronounced turbulence accompanying flash-back observed in this tube in this mixture range. Turbulence increases the burning velocity and hence the critical flow for flash-back. Although the Reynolds number for turbulence is not exceeded in the flow of unburned and burned gases, turbulence may be induced if the diffraction of the stream lines that takes place at the combustion zone exceeds some critical limit. The diffraction increases with increasing ratio of combustion surface to cross section of the tube, that is, with increasing tilt of the combustion zone in the tube. The cross

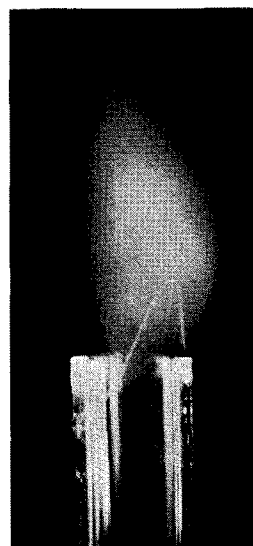


FIG. 10. Flame partly drawn into burner tube at gas flow just exceeding critical flow for flash-back.

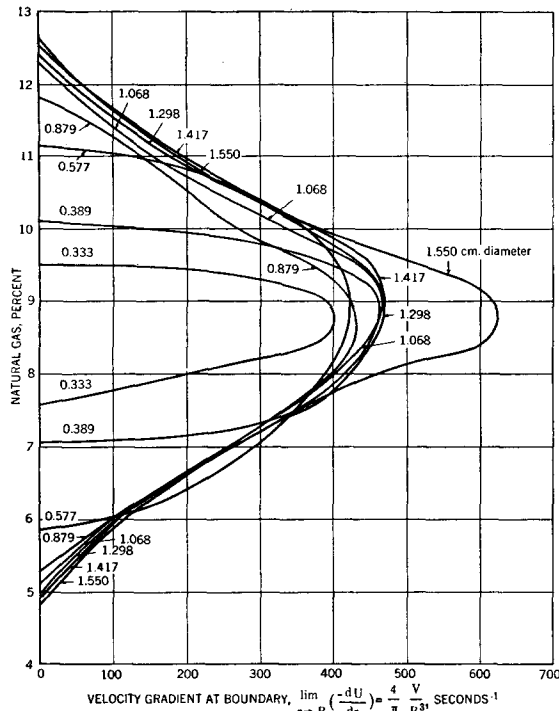


FIG. 11. Critical velocity gradients at boundary of gas stream for flash-back of natural gas-air flames in cylindrical tubes of different diameters at room temperature.

section increases with the square of the diameter; the combustion surface at the critical flow for flash-back, being approximately proportional to the flow, increases [according to Eq. (4)] with the cube of the diameter.

Figure 13 shows critical flows for various tube diameters above which flames burning in air are blown off. The slope of the curves changes markedly on approaching the condition of turbulent flow (Reynolds number > 2000). The junction of blow-off and flash-back curves is shown in detail in Fig. 14 for several tube diameters. For the larger tubes, the blow-off curve approaches the flash-back curve asymptotically. In smaller tubes a region of extinction appears at small flows. On approaching this region from the region of stable flame the effects of dilution and the rim increase, the point of equality of burning velocity and gas velocity shifts to the axis, and the flame assumes a water-meniscus-like shape. Finally, at the extinction limit the explosive reaction is quenched and the flame goes out. In a small tube where flash-back is possible, the latter can be forced at the appro-

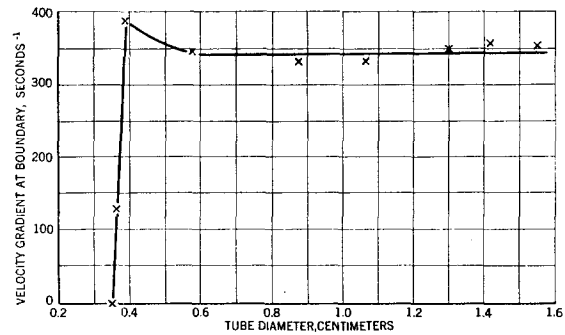


FIG. 12. Critical velocity gradients at boundary of gas stream for flash-back of flames of 7.5 percent natural gas-air mixtures in cylindrical tubes of different diameters at room temperature.

priate flow, either by heating the rim of the tube or, where extinction and flash-back curves are close together, by first forming a stable flame and suddenly reducing the flow to the appropriate value. Where flash-back and extinction curves are very close to each other, for example, at 9.04 percent natural gas in the 0.333-cm tube, the flame is curiously unstable, tending unpredictably to extinction or flash-back.

A plot of

$$\lim_{r \rightarrow R} \left(-\frac{dU}{dr} \right)$$

against mixture composition, calculated from Eq. (4) for the blow-off curves in Figs. 13 and 14, is shown in Fig. 15. The points are not drawn in because the agreement is so close, even for the blow-off curves for small tubes almost down to the junction with the extinction curves, as to be indistinguishable on the plot. Thus, for otherwise equal conditions, the velocity gradient at the boundary for blow-off is the same for all tubes. To Fig. 15 is added the average flash-back curve derived from Fig. 11 for medium-size tubes, thus providing a graph of the region of stable flames of natural gas for the condition of laminar flow.

As shown in Fig. 16, the blow-off condition changes profoundly with the nature of the surrounding atmosphere. If the surrounding atmosphere is oxygen, the region of stable flame greatly increases (see also Fig. 13); if an inert gas, the region is sharply limited, the blow-off closing in to the flash-back curves both on the lean and rich sides. The effect increases in the order nitrogen, carbon dioxide, helium. It

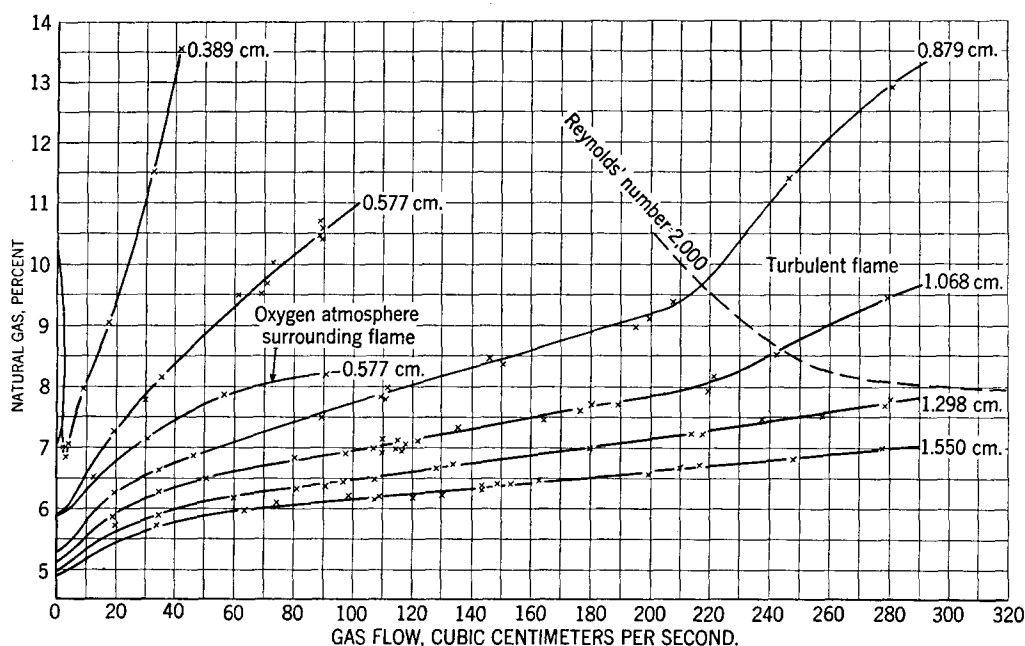


FIG. 13. Critical flows for blow-off of natural gas-air flames from cylindrical tubes of different diameters at room temperature. Flames surrounded by air except as indicated.

demonstrates the effect of interdiffusion on the burning velocity in the boundary layer of the stream. That the largest effect is obtained with a surrounding atmosphere of helium is understood from the larger diffusion coefficient of this gas. The greater effect of carbon dioxide as compared with nitrogen suggests the influence of the higher heat capacity of this gas on the temperature in the combustion zone at the boundary. It was found that the velocity of the surrounding gas parallel to the burner tube had no effect on the blow-off over a considerable range. This is understandable because the surrounding gas in contact with the flame gas at the burner rim is retarded by friction at the outer surface of the burner tube. Only if the gas is blown across the rim does its velocity affect the blow-off strongly.

Since quenching of the explosive reaction by the surface of the tube is attributed to cooling and destruction of chain carriers (atoms, free radicals), the question may be raised whether the nature of the surface of the burner tube influences the flash-back and limit of inflammability. Experiments with Pyrex tubes coated with salts such as sodium tungstate and potassium chloride and with silver showed the same limits and flash-backs as with clean Pyrex tubes.

This contrasts sharply with the effect of these surface treatments on reaction rates and explosion limits of hydrogen-oxygen and hydrocarbon-oxygen mixtures enclosed in heated vessels. In the latter experiments cooling effects are negligible because the reacting gas is virtually at the same temperature as the wall of the reaction vessel, and the quenching effect of surface is confined to the destruction of chain

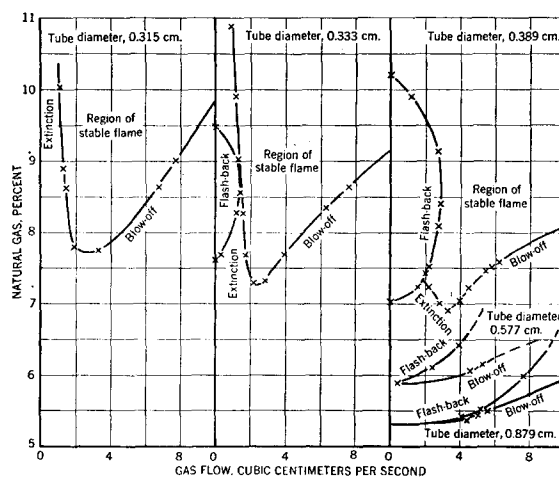


FIG. 14. Regions of stable natural gas-air flames in air for several cylindrical tubes at small flows.

carriers. In the present experiments, however, the cooling effect is very large. It is virtually independent of the nature of the tube material because the heat conductivity of a solid exceeds that of a gas by several orders of magnitude. The wall, therefore, constitutes an effective temperature-sink, and the rate of heat transfer is governed by the heat conductivity of the gas. Whether cooling overshadows destruction of chain carriers by the wall cannot be stated with certainty at present; the experiments do not allow a decision because destruction of chain carriers may also be independent of the nature of the surface for the following reason: In the combustion zone the concentrations of atoms and free radicals are very large, being comparable (within an order of magnitude) to the concentrations of the reactants themselves.⁹ Therefore, any surface close to the combustion zone is well-covered with adsorbed atoms and radicals, and such species diffusing to the surface easily find reaction partners for recombination to neutral molecules, the wall serving as third body. The rate of chain-carrier destruction, therefore, becomes equal to the rate of diffusion independent of the nature of the surface. This contrasts with the slow reaction between hydrogen

or hydrocarbons and oxygen, where the chain-carrier concentrations are very low; here the probability of recombination at the surface depends strongly on the adsorptivity of the surface. A clean Pyrex surface, for example, permits many chain carriers to return to the gas phase and therefore has a low chain-breaking efficiency. A salt-coated surface has a relatively large adsorptivity for chain carriers and therefore a high chain-breaking efficiency.¹⁰

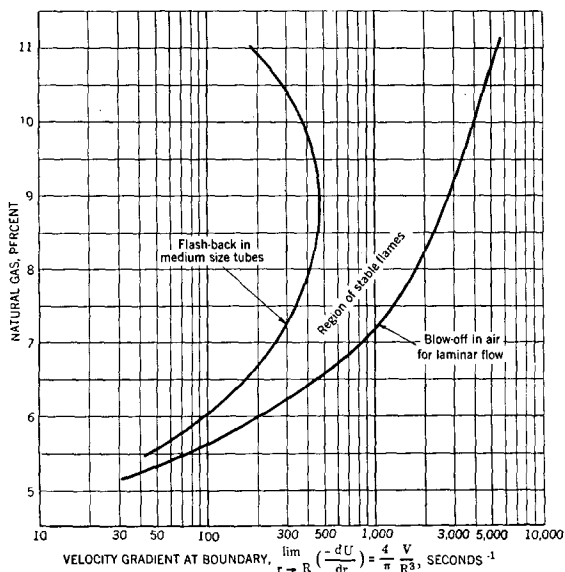


FIG. 15. Region of stable natural gas-air flames for laminar streams in cylindrical burner tubes.

⁹ B. Lewis and G. von Elbe, *J. Chem. Phys.* **2**, 537 (1934).

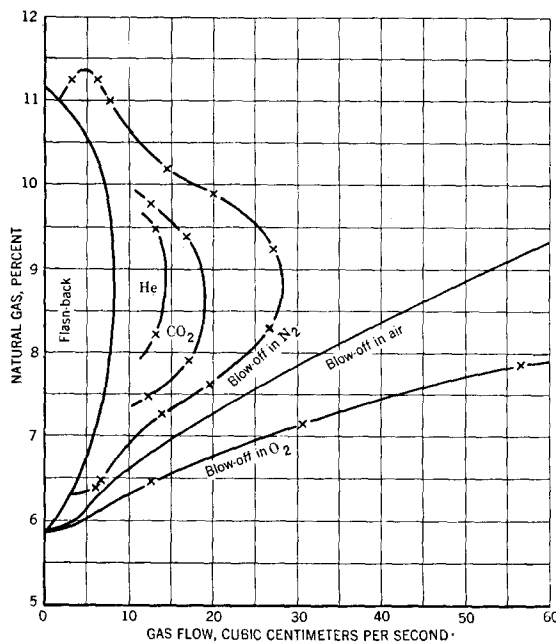


FIG. 16. Effect of nature of surrounding atmosphere on blow-off of natural gas-air flames from cylindrical burner tube. Tube diameter 0.577 cm.

(b) Inverted Flames

At flows exceeding the critical for blow-off, a zone of equality of burning and gas velocities may be formed above an obstruction in the stream by the effects of friction and quenching and may support a stable flame over some range of flows. If the obstruction consists of a wire mounted in the axis of a cylindrical burner tube, the outward propagation of the combustion zone from the point of equality above the wire and the upward flow of the gas results in an inverted cone, of which the photograph in Fig. 17 is an example. In taking this and similar photographs,

¹⁰ G. von Elbe and B. Lewis, *J. Chem. Phys.* **10**, 366 (1942).

the light beam was narrower than the wire to avoid illuminating particles in front of and behind it. Particles in the shadow of the wire are invisible. The inverted cone blows off above some critical flow. When the flow is decreased the tip approaches the wire and as the latter is heated its quenching effect may decrease sufficiently to permit the combustion zone to travel down its length within the adjacent slow-moving gas layer. The inverted flame is then drawn through the burner tube as successively lower parts of the wire are heated by conduction.

Figure 18 is a photograph of a dust-laden air stream from a cylindrical tube of 1.068-cm diameter containing a centrally mounted wire of 0.107-cm diameter. The measured velocities at various distances r from the axis of the tube, corrected for free fall of the particles, are plotted in Fig. 19. Close agreement is found with the velocities calculated from the equation of flow through an annular space,¹¹

$$U = n \left(R^2 - r^2 + \frac{R_w^2 - R^2}{\ln R_w/R} \ln r/R \right), \quad (5)$$

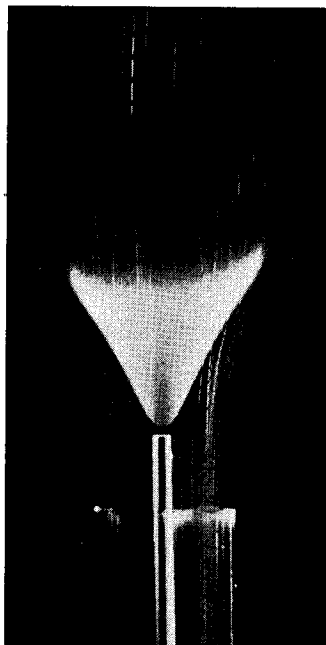


FIG. 17. Inverted flame cone with particle tracks. Tube diameter, 1.717 cm, wire diameter, 0.211 cm; mixture composition, 6.20 percent natural gas in air; gas flow, 168 cc/sec. Particles invisible in shadow of wire on the left.

¹¹ Lamb, *Hydrodynamics* (Cambridge Press, 1924), fifth edition, p. 555.

and

$$V = \frac{\pi}{2} n \left(R^4 - R_w^4 + \frac{(R_w^2 - R^2)^2}{\ln R_w/R} \right), \quad (6)$$

where R_w is the radius of the wire.

For a given wire the critical flows for blow-off of the inverted cone for various tubes are such that the velocity gradient at the surface of the wire remains constant (Figs. 20 and 21). The gradients are calculated from the following equations, which are obtained by differentiation of Eq. (5) and combining with Eq. (6),

$$\lim_{r \rightarrow R_w} \frac{dU}{dr} = n \left(-2R_w + \frac{R_w^2 - R^2}{\ln R_w/R} \frac{1}{R_w} \right), \quad (7)$$

$$= \frac{2}{\pi} \frac{V}{R^3} \frac{b \left(\frac{1 - 1/b^2}{\ln b} - 2 \right)}{1 - b^4 + \frac{(b^2 - 1)^2}{\ln b}}, \quad (7a)$$

where $b = R_w/R$.

If the diameter of the wire is varied for a given tube diameter (Fig. 22), the critical velocity gradient remains constant over a considerable range of wire diameters, as shown in Fig. 23. For the smallest wire (0.0643 cm) a slight decrease and for the largest wire (0.536 cm) a considerable increase are observed. The decrease toward smaller wires is explained by the decrease of the burning velocity at the tip of the inverted cone:

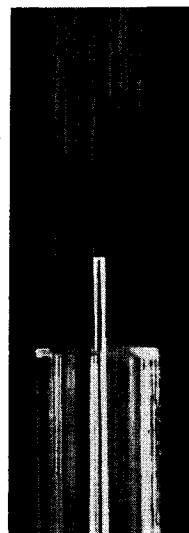


FIG. 18. Particle tracks in an air stream from a cylindrical tube of 1.068-cm diameter with central wire of 0.107-cm diameter. Gas flow, 155 cc/sec.

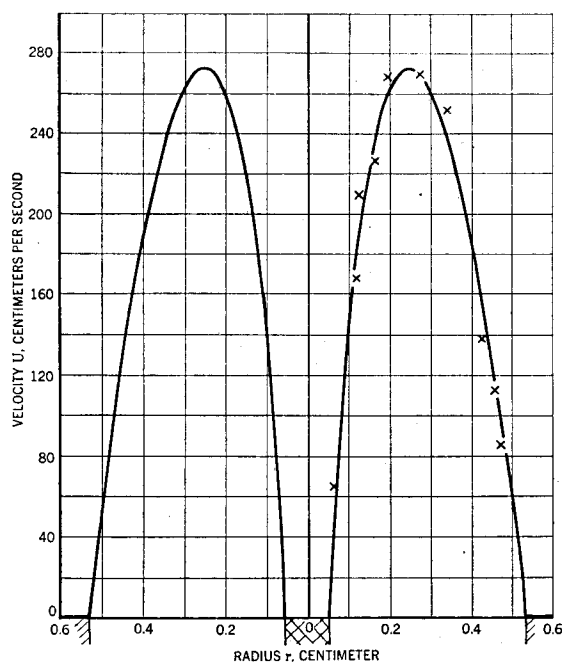


FIG. 19. Velocity distribution in a laminar stream of air (155 cc per second) at the orifice of a cylindrical tube of 1.068-cm diameter, in whose axis a wire of 0.107-cm diameter is mounted. Temperature 25°C; pressure 740 mm Hg. $\times$, experimental velocity. —, calculated velocity.

the stream of slow-moving gas above a fine wire is confined to such a fine filament that the combustion zone at the tip of the inverted cone tends to become highly curved and therefore loses more heat and chain carriers to the unburned gas than the relatively flatter cone tip above a larger wire. For very large wires there is an appreciable decrease of gas velocity above the wire because, as the flow lines close in toward the axis (as may be observed in the example of Fig. 18), the cross section of the stream is increased. Large wires therefore retain the flame better.

Comparison of velocity gradients for blow-off from wires with those for blow-off of upright flames in air shows that the former are much larger, due to absence of the diluent effect of the surrounding atmosphere; this fact explains why a Méker burner may operate on leaner mixtures than the ordinary Bunsen burner.

3. FLOW PATTERN AND TEMPERATURE DISTRIBUTION IN THE FLAME

(a) Upright Flames

The diagram in Fig. 24, which was drawn from photographic records of flames obtained under

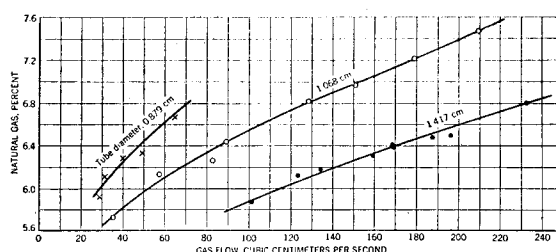


FIG. 20. Critical flows for blow-off of inverted natural gas-air flame cones from end of a wire mounted in axes of cylindrical tubes of various diameters. Diameter of wire 0.107 cm.

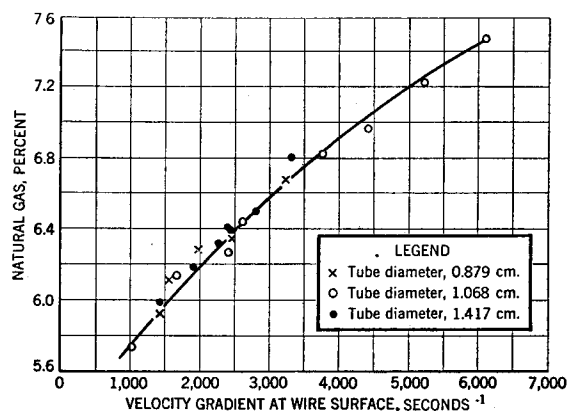


FIG. 21. Critical velocity gradients at wire surface for blow-off of inverted natural gas-air flame cones from end of a wire mounted in axes of cylindrical tubes of various diameters. Diameter of wire 0.107 cm.

the same conditions as Fig. 1, illustrates the flow pattern of the laminar gas stream in the flame. The combustion zone forms an angle with the flow lines so that the normal component of the gas velocity equals the burning velocity. Thermal expansion increases the gas velocity as well as the angle of the flow lines in such a way that the mass flow in any filament remains constant. It is observed that the refraction of a flow line begins well before it enters the luminous combustion zone, indicating penetration of heat into the unburned gas to a considerable depth. This effect gives rise to schlieren, or striations, in front of the combustion zone which are particularly well observed with inverted flames because there the burned gas is not in the line of sight. It is also strikingly demonstrated in Fig. 25 where stannic chloride hydrate smoke has been added to an inverted flame. The smoke disappears on approaching the combustion zone owing to dissociation of the hydrate, the zone of disappear-

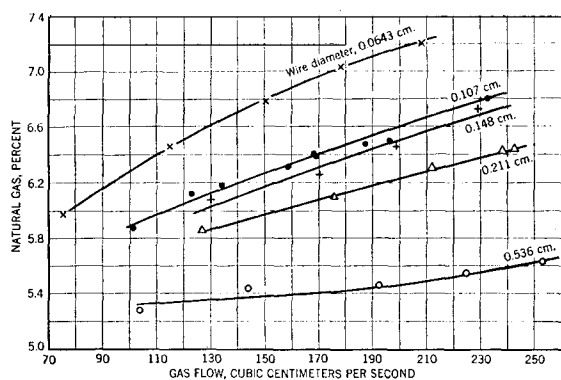


FIG. 22. Critical flows for blow-off of inverted natural gas-air flame cones from end of wires mounted in axis of a cylindrical tube of 1.417-cm diameter.

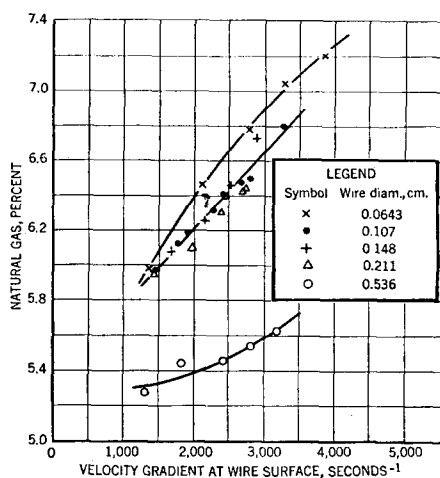


FIG. 23. Critical velocity gradients at surfaces of wires for blow-off of inverted natural gas-air flame cones from ends of wires mounted in axis of a cylindrical tube of 1.417-cm diameter.

ance presumably beginning at a temperature close to 83°C at which the tri-hydrate is known to decompose. Furthermore, Fig. 24 shows that the outer flow lines deflect away from the axis considerably before they reach the zone of heat penetration; this is attributable to back pressure caused by acceleration of expanding gas in the combustion zone, forcing the stream to diverge as it emerges from the orifice. The divergence is particularly noticeable in flames of low gas velocity, of which Fig. 26 is an example. The outermost flow lines near the rim bend around sharply owing to heat as well as back pressure. In consequence, combustible gas is carried beyond the radius of the tube, causing the base of the cone to extend beyond the orifice and

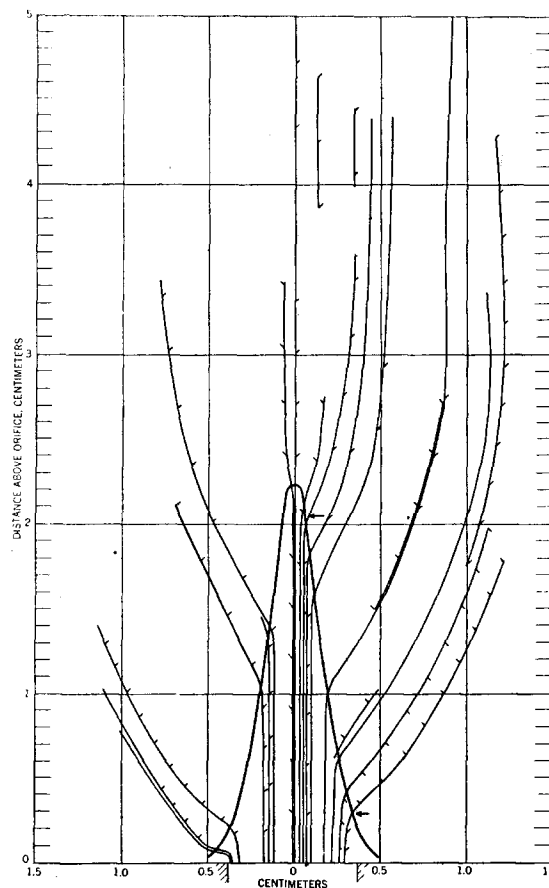


FIG. 24. Diagrammatic reproduction of the flame and particle tracks in Fig. 1 and additional photographs. Rectangular burner tube, 0.755×2.19 cm. 204 cc per second, 7.50 percent natural gas in air. Cone outline is the inner border of the luminous combustion zone of 0.2- to 0.3-mm thickness. Arrows indicate range of constant burning velocity. Particle tracks are interrupted in intervals of 1.436×10^{-3} second.

forming the overhang commonly observed. Following the combustion zone downward, one observes that the angle between flow lines and combustion zone increases until a flow line enters it at right angles; here the gas velocity, modified by thermal and dynamic effects, and burning velocity, modified by the rim effect and interdiffusion with the surrounding atmosphere, are equal. Below this point the gas velocity is again larger than the burning velocity, as may be seen from the fact that the angles between flow lines and combustion zone are not right angles. This corresponds to the illustration in Fig. 7b. The lowest visible flow lines are seen to skirt the luminous zone, and the gas carried up in the

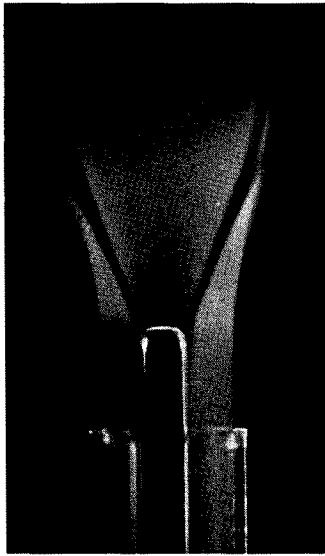


FIG. 25. Penetration of heat from combustion zone into unburned gas made visible by disappearance of stannic chloride hydrate smoke.

outermost layers reacts comparatively slowly downstream in the envelope.

The burning velocity S_u , that is, the rate of progress of the combustion zone in the normal direction, is defined by the equation of conservation of mass:

$$S_u dl = U_u dr, \quad (8)$$

where U_u is the velocity of the unburned gas in a filament of width dr , and dl is the width of the same filament as it enters the combustion zone. From a diagram of the type of Fig. 24, the length l of the cone outlined from the tip to the intersection of any of the registered flow lines originating at various radii r at the orifice may be determined, and from the plot of l versus r the differential quotient dl/dr for any value of r may be obtained. As U_u is also known as a function of r [Eq. (1)], S_u can be determined over the entire cross section of the stream from Eq. (8). Figure 27 shows burning velocities over the cross section

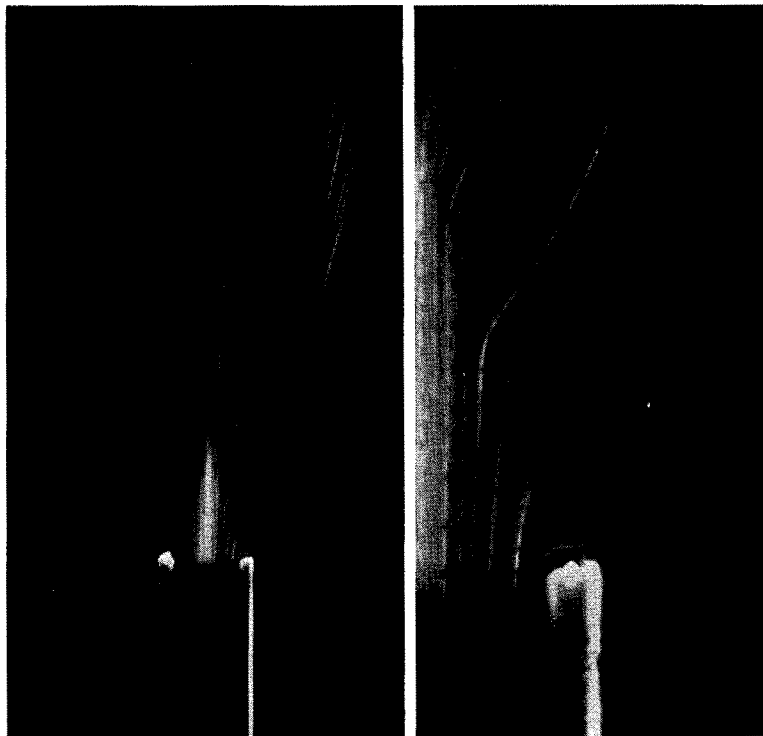


FIG. 26. Flame of low gas velocity, showing divergence of unburned stream caused by back pressure. Enlargement shows outermost flow lines skirting base of cone. Rectangular burner tube; gas flow, 88 cc/sec.; mixture composition, 6.32 percent natural gas in air.

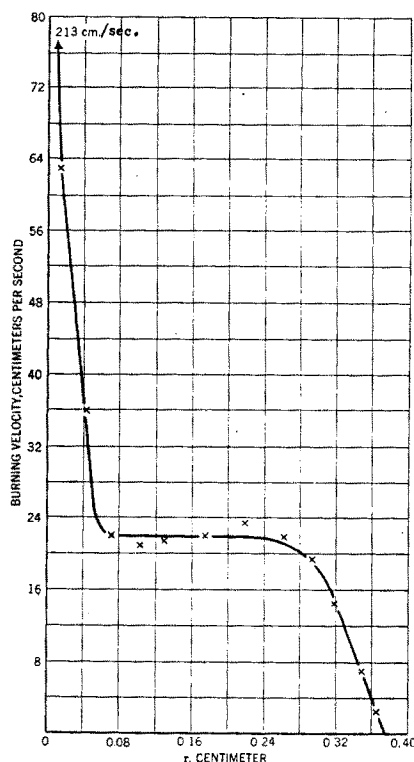


FIG. 27. Burning velocity of flame (Fig. 24) in the flow lines originating at distance r from axis of burner tube. Points correspond to readings of dl/dr from plot of l versus r (see text).

of the stream for the flame illustrated in Fig. 24. The burning velocity is highest at the rounded tip where the axial flow line enters the combustion zone in the normal direction, namely $S_u = U_u = 213$ cm per second. From the tip downward, the burning velocity decreases sharply to a constant value of 22.0 cm per second, which extends over the length of flame outline indicated by arrows in Fig. 24, and then drops to zero on approaching the base. The high value at the tip reflects the convergence of heat flow and chain-carrier diffusion into the narrow unburned stream from the adjacent burning gas.¹² At the

¹² The increase in burning velocity from this cause also may explain the formation on cylindrical tubes of Bunsen flames of polyhedral structure described by F. A. Smith and S. F. Pickering [Bur. Stand. J. Research 3, 65 (1929)]. These authors found that under certain conditions a circular Bunsen cone changed to a polyhedral cone having from 3 to 7 sides and the appearance of a "tent supported by ropes loosely draped from a center pole to pegs located at the corners of a regular polygon." The figure can be made to remain stationary or to rotate by changing very slightly the composition or velocity of the gas mixture. As the burning velocity is obviously larger along the ridges than along the

base, besides the rim effect and interdiffusion with the surrounding atmosphere, the divergence of heat flow and chain-carrier diffusion contributes to the low burning velocity.

The usual procedure for determining burning velocity has been to divide the flow by the cone area, which provides an average value over the whole flame. The corresponding procedure—namely, dividing the flow in the plane shown in Fig. 24 by the total length of the cone outline—yields the average burning velocity of 23.4 cm per second in agreement with the above constant value, showing that the high and low burning velocities around the tip and the base, respectively, approximately cancel each other. It is clear that such agreement is conditioned by the choice of orifice dimension and gas flow and cannot be expected as a general rule. If, for example, the orifice were smaller, the cone height would, for the same average gas velocity, decrease approximately proportionately, while the height of the disturbed zone from the base upward would remain approximately constant and contribute proportionately more to depressing the average burning velocity. Corresponding considerations apply to circular burners, and indeed Smith⁶ found that the average burning velocity obtained from cone area and flow decreases with decreasing tube diameter. Smith

planes of the polyhedron, the equality of gas and burning velocity is established at the end of the ridges at the base; the flame is, therefore, stabilized only at discrete points above the rim. A steady state requires that the angle between the faces of the polyhedron which is necessary to provide a burning velocity equal to the gas velocity be that of a regular polygon, because only in this way can dynamic symmetry be established. When the angle is not that of a regular polygon, this symmetry is destroyed and the figure rotates. Consequently, it is expected that for otherwise constant conditions a gradual change of mixture composition toward lower burning velocities should produce successive transitions from a circular cone to many-sided, down to three-sided polyhedrons and finally blow-off, the decreasing angle between the faces compensating for the loss of burning velocity on changing the mixture. Unfortunately, the experiments have not been performed in this way. Smith and Pickering varied the composition and gas flow simultaneously. The phenomena are readily observed with rich flames of the heavier hydrocarbons surrounded by a diluent gas, as is the case in a Smithells separator. The susceptibility of the heavier hydrocarbons to formation of polyhedral flames may be ascribed to the slowness of the explosive reaction with the consequent relatively large loss of heat and chain carriers to the preheated zone. If this loss is diminished by the approach of another combustion zone a large gain in reaction rate results. If the flame is surrounded by air then obviously the increased burning velocity at the base caused by interdiffusion of secondary air always preserves a circular cone.

himself determined the burning velocity from the angle of the cone surface at the distance $0.707 R$, the supposition being that the cone surface at this point is intersected by a flow line in which the gas velocity is the average velocity over the cross section. It is true that within the tube the average gas velocity is at $0.707 R$, but, as seen from Fig. 24; the flow lines bend outward above the orifice and at $0.707 R$ the cone surface is intersected by a more central flow line. This introduces a methodical error into Smith's method which, however, is not serious as long as the cone angle does not greatly change in the neighborhood of the point of measurement. Another criticism¹³ of Smith's method, namely, that the velocity distribution over the cross section changes above the orifice, is invalidated by the present experiments, which show this effect to be negligible for usual cone heights.

The example of Figs. 24 and 27 is typical for flames of all mixture compositions outside of the very rich mixture range where the combustion zone becomes diffuse and is eventually obscured by the formation of free carbon (yellow tip) and combustion in secondary air. Several flames of different mixture compositions have been analyzed and the burning velocities in the constant range plotted in Fig. 28. The burning velocity is affected by the introduction of fine dust in considerable and uncontrolled quantity together with coarser particles. No effort was made to eliminate this source of error, since determination of burning velocity was not the primary purpose of this work, and the results, therefore, show considerable scattering.

In the course of these experiments it was noted that the introduction of stannic chloride hydrate smoke markedly lowered the burning velocity and, correspondingly, the critical flow for blow-off. That the inhibition of the explosive reaction was not caused by hydrochloric acid set free in the hydrolysis of stannic chloride was shown by the ineffectiveness of hydrochloric acid gas prepared from sulphuric acid and sodium chloride. The smoke gives rise to incandescence at the combustion zone, reminiscent of soot-forming flames, and the formation of particles of stannic oxide.

¹³ Reference 5, page 86.

From knowledge of the burning velocity and the flow pattern it appears possible, in principle, to calculate the form of the inner cone from the condition that at the cone surface the normal component of the gas velocity equals the burning velocity, that is:

$$S_u = U_u \sin \alpha, \quad (9)$$

where α is the angle of the cone surface with the flow line on the unburned side. It is not possible, however, to treat this problem in closed mathematical form without simplifying assumptions because of disturbances at the tip and base of the cone, the finite thickness of the combustion zone, and the outward deflection of the flow lines caused by back pressure. Michelson's¹⁴ treatment makes the following simplifying assumptions:

- The burning velocity is constant over the whole cone surface.
- The boundary between unburned and burned gases approximates a mathematical surface, the temperature changing abruptly from the initial to the final on passing through it.
- The flow lines retain their direction and velocity from the orifice right up to the cone surface.

Despite obvious limitations imposed by these assumptions, it is instructive to compare the cone surface calculated from Michelson's equation with the actual. In the following calculation Eq. (1) is used for the velocity distribution in

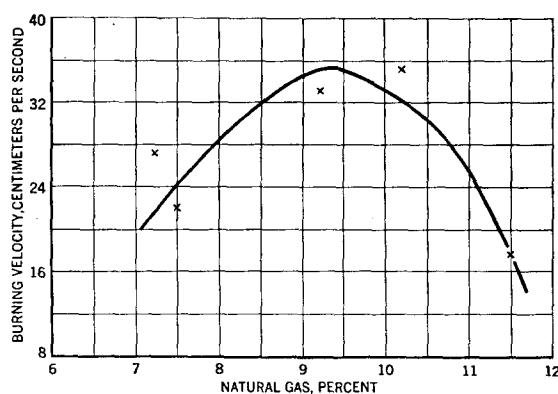


FIG. 28. Burning velocities in the constant range between tip and base of flame cones for various mixtures of natural gas and air.

¹⁴ Cf. reference 4, page 48.

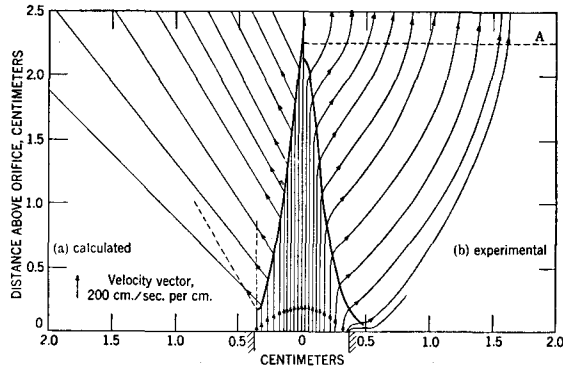


FIG. 29. Flame cone and flow pattern from analysis of the flame of Fig. 24.

the unburned gas; this differs from Michelson's treatment in which a finite gas velocity at the boundary exceeding the burning velocity was assumed, contrary to actual conditions. By retaining assumption (a) and using Eq. (1) it is evident that the cone surface can only be followed to a point near the boundary where the gas velocity equals the assumed constant burning velocity. The latter exceeds the gas velocity beyond this point, and no real solution to the equation of the cone surface exists. This is unimportant, however, because it concerns the disturbed boundary region where the theory is not applicable. The treatment yields only the shape of the cone and does not define its position with respect to the orifice. It may be mentioned that this is not remedied by Michelson's assumption of a finite gas velocity at the boundary exceeding the burning velocity. That is, Michelson's theoretical flame would blow off while the present theoretical flame would flash back.

The equation for coordinates r and z for any point on the cone surface, where r is the distance from the axis and z the height of the cone tip above the point, was obtained by Michelson by integrating the equation

$$dz/dr = \cot \alpha, \quad (10)$$

which, by means of Eq. (9), transforms to

$$\frac{dz}{dr} = \pm \frac{(U_u^2 - S_u^2)^{1/2}}{S_u}, \quad (11)$$

where U_u as a function of r is given by Eq. (1),

and S_u is constant. The result is:

$$z = \pm \frac{M}{3} \left[\sin \varphi \cos \varphi \Delta \varphi + \left(1 - \frac{1}{k^2} \right) F(k, \varphi) + \left(1 + \frac{1}{k^2} \right) E(k, \varphi) \right], \quad (12)$$

where

$$M = [nR^2/S_u - 1][(nR^2 + S_u)/n]^{1/2},$$

$$k^2 = (nR^2 - S_u)/(nR^2 + S_u),$$

$$\sin^2 \varphi = nr^2/(nR^2 - S_u),$$

$$\Delta \varphi = (1 - k^2 \sin^2 \varphi)^{1/2},$$

and $F(k, \varphi)$ and $E(k, \varphi)$ are elliptical integrals of the first and second kinds,

$$\int_0^\varphi \frac{d\varphi}{\Delta \varphi} \quad \text{and} \quad \int_0^\varphi \Delta \varphi d\varphi,$$

respectively. $n = -a/4\eta$ as above. Depending on the sign, the cone is upright or inverted. It can be shown theoretically that the upright cone is the stable form when the base is fixed in the stream and the inverted cone is the stable form when the tip is fixed in the stream.¹⁵

Figure 29a shows a cone calculated from Eq. (12) for the flow and burning velocity (22 cm per second) of the flame of Fig. 24. In Fig. 29b the experimental cone of Fig. 24 is reproduced with an interpolated set of flow lines. The calculated cone is placed in such a position above

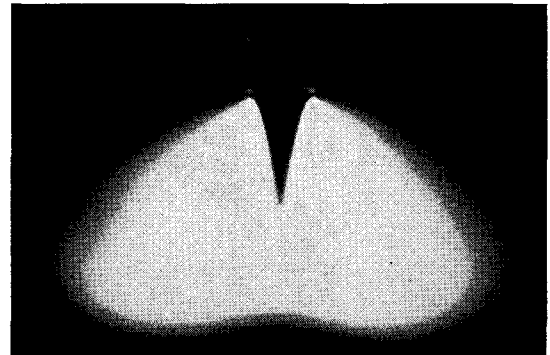


FIG. 30. Flame from rectangular burner tube, directed downward to accentuate separation of flame envelope. Sodium-colored for photographic contrast. Mixture composition, 8.46 percent natural gas in air; gas flow, 185 cc/sec.

¹⁵ Reference 4, page 51.

the orifice that the flow line at $r=0.147$ cm intersects the cone outline at the same height above the orifice as the corresponding flow line of the experimental cone outline, namely, at 1.10 cm. It is seen that over the range of constant burning velocity the heights of the intersections of all other flow lines agree well in experiment and theory. Toward the base and the tip there occur the discrepancies expected from the decrease and increase of S_u , respectively; that is, toward the base the calculated cone outline is too short and toward the tip too long. The calculated outline ends at $r=0.357$ cm, where $U_u=S_u$, that is, 0.02 cm from the stream boundary. At this point the cone surface is highly curved, in itself suggesting a physical instability; that is, the burning velocity should decrease because of the extreme divergence of heat flow and diffusion in the direction of burning gas to unburned gas. Some curvature is preserved at the base of the actual cone, and this may contribute to the decrease of burning velocity toward the base. The sharp tip of the theoretical cone is evidently physically impossible; as mentioned before, preheating and preactivation of the unburned gas in this region from the adjacent combustion zone transform this to a rounded tip in the actual flame.

The theoretical flow lines are continued in the burned gas in Fig. 29a by means of the relation

$$\rho_b U_b \sin \beta = \rho_u U_u \sin \alpha, \quad (13)$$

which equates the components of mass flow normal to the cone surface. ρ_u and ρ_b are the densities of unburned and burned gases, respectively, and β is the angle between flow line in the burned gas and cone surface. Assumption (b) above leads, furthermore, to the relation

$$U_b \cos \beta = U_u \cos \alpha, \quad (14)$$

and hence

$$\rho_u / \rho_b = \tan \beta / \tan \alpha. \quad (15)$$

ρ_u / ρ_b is taken as 5.90, corresponding to the flame temperature of 1750°C (see below), the initial temperature of 27°C and a change in molar volume of 1½ percent. The theory is inadequate for the description of convergent flow at the highly curved base, but this problem has no interest, inasmuch as the calculated base is unreal. It is seen that on passing through the

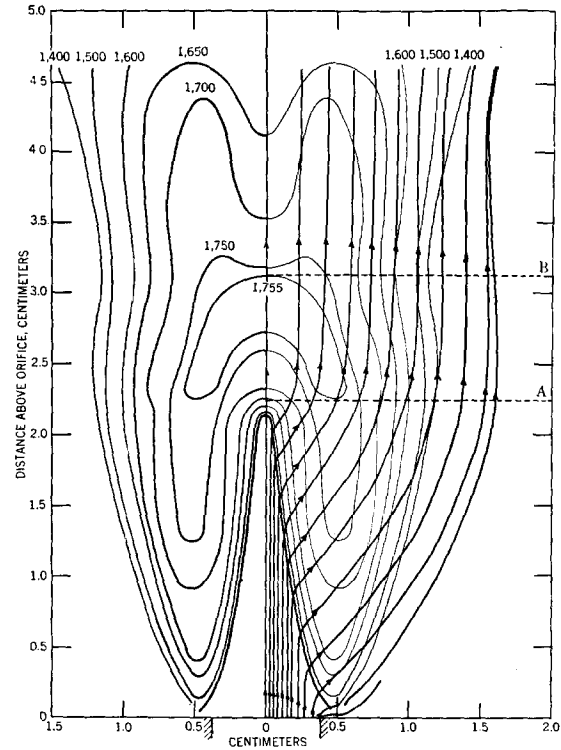


FIG. 31. Temperature distribution and flow pattern in the flame of Fig. 24. Temperatures in °C.

combustion zone the gas acquires a substantial velocity normal to the axis. The resulting divergence of the stream causes the maximum gas velocity to shift from the central flow line outward, as is shown by the actual flow pattern in Fig. 29b (compare velocity vectors at plane A). If the burner tube were circular, the horizontal velocity components along any flow line would decrease proportional to the distance from the axis because of the requirement of constancy of mass flow while the velocity components parallel to the axis remain constant (except for secondary effects of cooling, friction, and convective rise). The flow lines, therefore, would bend upward. In a burner tube of narrow rectangular cross section used in the present experiments the flow lines retain their direction, except as modified by convective rise of the hot gas, because the area through which the burned gas passes remains constant with distance from the axis. In this burner the flame envelope, therefore, splits into two parts. This is seen, for example, in Fig. 4. The separation would be greater were it not for convective rise. If the burner tube is turned

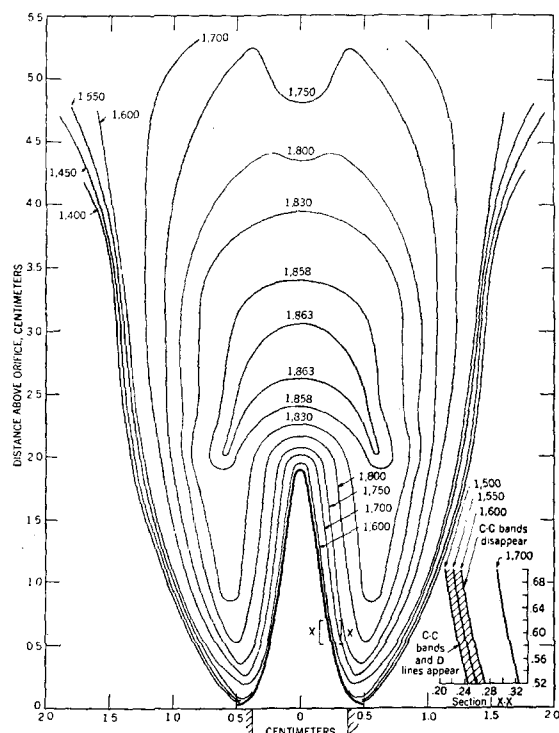


FIG. 32. Temperature distribution in flame of 8.70 percent natural gas in air; 224 cc per second; rectangular burner tube, 0.755×2.19 cm. Temperatures in °C.

upside down, convective rise accentuates the separation as shown in Fig. 30.

Because of assumption (b), the refraction of the flow lines in the theoretical flame is discontinuous. In the actual flame, where the temperature rise is spread out over some distance (see Fig. 31), the refraction is gradual. Nevertheless, the refraction angles agree rather well in experiment and theory, except in the neighborhood of the tip and base of the cone, until convective rise becomes effective. Similarly, the calculated velocities U_b [from Eqs. (14) and (15)] agree with the experimental U_b 's except in the neighborhood of the tip, where the gas in the actual flame passes through a much smaller cone surface; hence U_b is much larger than calculated.

The flow lines in Fig. 29b were corrected for inertia of the particles, which tend to retain their original direction of motion and not to follow completely the change in direction of the gas. There is evidence of inertia from occasional photographic tracks that show unusually small refraction angles. Furthermore, for a number of tracks it was found that the condition of con-

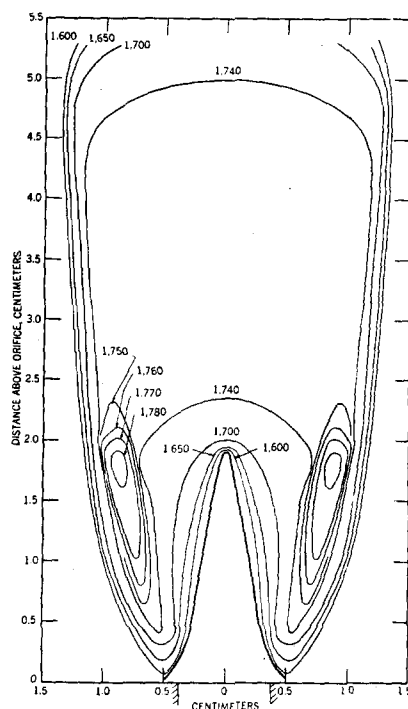


FIG. 33. Temperature distribution in flame of 10.96 percent natural gas in air; 161 cc per second; rectangular burner tube, 0.755×2.19 cm. Temperatures in °C.

stancy of mass flow was not fulfilled. This condition requires that the area between the velocity vectors of two neighboring flow lines in the burned gas should be ρ_u/ρ_b times the corresponding area in the unburned gas. Particularly at the tip, particles in the burned gas are generally too close together. This region is marked by considerable divergence of the flow lines, and hence the inertia effect is most pronounced here. On the side of the cone, especially in the region of constant burning velocity where the flow lines are approximately parallel, tracks of two adjacent particles are about equally affected by inertia, and the error tends to cancel out. In this region the ratio of the areas between the velocity vectors agrees with the theoretical value of ρ_u/ρ_b , increasing from the combustion zone outward in correspondence with the temperature rise.

Correction of the flow lines in Fig. 29b for particle inertia was made in the following way. Except in the immediate neighborhood of the combustion zone, particles may be supposed to follow the gas flow closely; from Fig. 24 the

velocity and direction of the gas at any point in the plane A above the tip may be obtained. If r_b is the distance of a point in plane A from the axis and $U_{b(n)}$ is the velocity component normal to the plane, the condition of constancy of mass flow in burned and unburned gases is

$$\rho_b U_{b(n)} dr_b = \rho_u U_u dr. \quad (16)$$

ρ_u/ρ_b at any r_b is obtained from the experimental temperature-distribution chart of Fig. 31. A curve of r_b versus r is constructed so that for any r_b differential quotient dr_b/dr equals $(U_u/U_{b(n)})(\rho_u/\rho_b)$. Thus, the intersection of any flow line originating at r in the plane of the orifice with plane A is known. To construct the flow lines, it suffices to determine the curve of r_b versus r in the plane A only, because for points not too close to the axis the intersections with plane A differ little from the photographed particle tracks, and near the axis (where the particle tracks are least reliable) the curved part of the flow line between r and r_b extends over only a short distance and cannot be appreciably in error.

Figure 31 shows the temperature distribution in the flame of Fig. 24. The corrected flow pattern of Fig. 29b is superimposed and extended with the aid of Fig. 24. The maximum temperature of 1755°C agrees with the temperature calculated for conditions of adiabatic combustion and thermodynamic equilibrium. It is attained in the protected central portion of the flame in about 0.0025 second after the gas has passed through the luminous combustion zone, although the main temperature rise, say to 1650°C , is completed in about 0.0005 second. Near the base, the cooling effect of the rim and dilution with the surrounding atmosphere is reflected in a sharp decrease of the temperature and, as is seen in Fig. 27, of the burning velocity. The temperature distribution in the envelope above the cone is conditioned by heat loss to the atmosphere, not only in the plane of the diagram but on all sides of the flame, because the envelope expands in the plane of the diagram and contracts in the plane normal to it, as may be seen in Fig. 4. The closed isotherms thus develop two maxima corresponding to the regions of highest flow (compare velocity vectors in plane B), since the slower-moving gas in the center, although protected by

regions of higher temperature in the plane of the diagram, loses heat in the direction normal to the plane. The progress of cooling of the flame gas is further recorded in the pattern of the outer isotherms which move first inward, coincident with the upward-bending of the flow lines, and then outward again as the heat flow from the interior of the stream becomes noticeable.

Figure 32 shows the isotherms for a near-stoichiometric flame. The temperature distribution is similar to that of Fig. 31. A detail (see insert) of the region near the luminous combustion zone is also given. In natural gas-air mixtures the luminous zone is marked by the sharp appearance and disappearance of the C-C (Swan) bands and C-H bands. The sodium D lines appear simultaneously with these bands, so that temperature readings can be taken within the combustion zone itself. As seen from the inset, the luminous combustion zone is about 0.2 mm thick. At the inner boundary of the zone where the D lines first appear the temperature has already risen to 1500°C , and at the outer boundary of

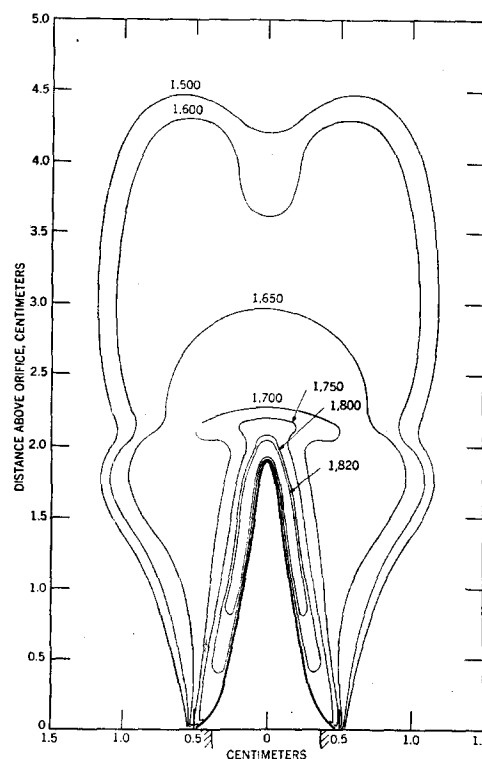


FIG. 34. Temperature distribution in flame of 7.06 percent natural gas in oxygen; 256 cc per second; rectangular burner tube, 0.755×2.19 cm. Temperature in $^\circ\text{C}$.

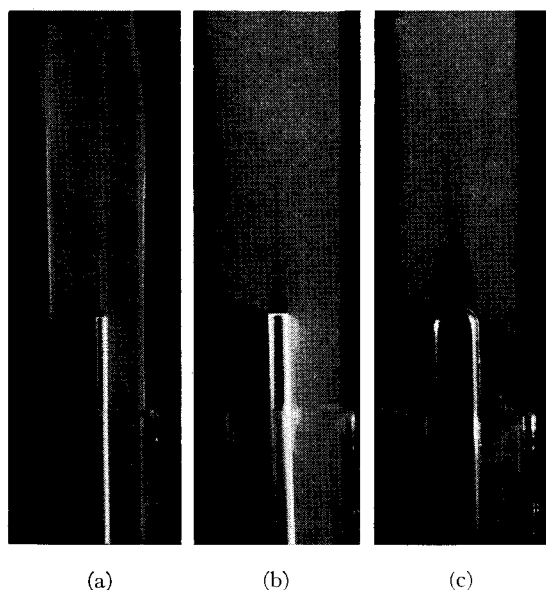


FIG. 35. Air streams with smoke from tubes with centrally mounted wires, showing annular vortex above wires.

	(a)	(b)	(c)
Tube diameter, cm	1.068	1.417	1.417
Wire diameter, cm	0.107	0.211	0.536
Gas flow, cc/sec.	155	189	168
Velocity gradient at wire surface, sec. ⁻¹	4500	2130	2130

the zone it has reached 1600°C. As the *D* lines are not observable in front of the inner boundary, the temperature gradient at the beginning of the combustion zone must be exceedingly steep.¹⁶

Figure 33 shows the isotherms for a very rich flame. Here the temperature distribution is modified by combustion with secondary air, hence the upper part represents a broad region of more or less constant temperature. The effect of secondary combustion is noted in the inward bending of the isotherms toward the base of the cone. This causes an increase in the burning velocity just above the base of the cone in agreement with the high gas flow required to blow off flames of rich mixtures. An interesting feature is the concentration of high temperature within the closed isotherms at the sides.

Figure 34 shows the isotherms of a lean natural gas-oxygen flame. Such flames differ from air mixtures, in that the highest temperature is found to border immediately on the luminous

combustion zone. Here *D*-line emission is exceptionally brilliant. In the luminous combustion zone no bands of C—C and C—H appear, the radicals evidently reacting rapidly with oxygen; only a continuous spectrum is visible, which also appears in the combustion zone of air flames but much weaker. *D*-line emission in the envelope is not as strong as in air flames, for the same NaCl concentration, although the temperature may be the same. This points to a decrease of the equilibrium concentration of sodium atoms on increasing the oxygen concentration. In accord with this is the observation that the most brilliantly colored flames are those that do not contain excess oxygen. The validity of the line-reversal method is independent of the sodium-atom concentration, the advantage of high concentration being the ease of determining the point of reversal. The maximum temperature in the flame of Fig. 34 exceeds the theoretical temperature by about 120°C.¹⁷ A discussion of this effect, for which there is also evidence from explosion-pressure measurements in excess oxygen mixtures, has been given elsewhere upon the basis of time lag of vibrational excitations.¹⁸

The difference between oxygen- and air-flames evidently is that in the former the combustion reaction is virtually completed in the combustion zone. A possible explanation is the burning of carbon monoxide which survives the reactions within the zone.

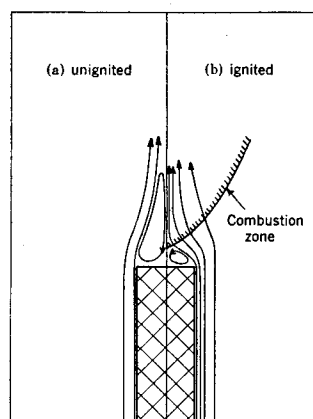


FIG. 36. Stream lines above wire with flat end, showing vortex. Wire diameter, 0.211 cm.

¹⁶ See also F. A. Smith and S. F. Pickering, *J. Res. Nat. Bur. Stand.* **17**, 7 (1936).

¹⁷ See also H. H. Kaveler and B. Lewis, *Chem. Rev.* **21**, 421 (1937).

¹⁸ Reference 7, pp. 306–344.

(b) Inverted Flames

The flow pattern of an inverted flame is illustrated by Fig. 17. The flow lines in the unburned gas are deflected outward by back pressure of the expanding flame gas and become parallel again on passing through the combustion zone. At the base of the cone where, in this example, the gas stream becomes turbulent, the combustion zone is always observed to trail because of the decreased burning velocity in the boundary of the stream. In the stagnant space above the wire, an annular vortex is formed that can be made visible with smoke in an air stream. Three such photographs are shown in Fig. 35. The cross section of the vortex is oval because the upward velocity of the outer filaments, which is imparted to them by the adjacent flow lines, exceeds the horizontal velocity of the filaments near the wire, which is decreased by friction. The vortex seems to be unimportant for the attachment of the flame, because it was found

that the critical flow for blow-off did not change when the end of the wire was rounded. The presence of the flame, however, greatly reduces the diameter of the vortex as is seen by comparing Fig. 35b with Fig. 17. In the latter the flame tip has compressed the region of vorticity. The original negative of the photograph in Fig. 17 shows particle tracks in vortex motion hugging the end of the wire below the combustion zone. Combustible mixture feeding the combustion zone at the tip is provided by laminar filaments above the vortex. This is illustrated in Fig. 36, which is a sketch of flow lines in the unignited and ignited streams.

Although the experiments described in this paper were performed with natural gas, which has a conveniently low burning velocity for studies in the laminar flow range and satisfactory activity for photographic exposures, it is evident that the mechanism of flame stabilization and the factors determining flame structure are equally operative for any fuel gas.

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The Ratio Rule

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An empirical rule for calculating the frequencies of isotopic groups in molecules is proposed and experimental evidence for it reported.

THE application of the product rule¹ is extremely useful for obtaining an approximate assignment of the fundamental frequencies of isotopic molecules. In the following a rule connected with the product rule will be described.

In the investigation of the isotopic methyl halides it was observed that each frequency of CD_3Cl could be predicted separately, in terms of the corresponding frequency of CD_3X , CH_3X ,

and CH_3Cl by means of the relation

$$\nu_{n\text{CD}_3\text{Cl}} = \frac{\nu_{n\text{CD}_3\text{X}}}{\nu_{n\text{CH}_3\text{X}}} \nu_{n\text{CH}_3\text{Cl}},$$

where for ν_n the corresponding frequencies have to be substituted. Table I gives the fundamental frequencies of CD_3Cl ² calculated by this rule from the CH_3Cl frequencies and those of the isotopic bromides.

Tables II-V give the calculated frequencies for

¹ Teller, cf. C. K. Ingold et al., *J. Chem. Soc. London*, 971 (1936); O. Redlich, *Zeits. f. physik. Chemie* **B28**, 371 (1935); E. B. Wilson, Jr., *J. Chem. Phys.* **9**, 76 (1941).

² H. D. Noether, *J. Chem. Phys.* **10**, 664 (1942).